DEPI GRADUATION PROJECT DOCUMENTATION

Manufacturing Line Productivity Analysis

1. Project Planning & Management | 2. Literature Review | 3. Requirements Gathering

1. Project Planning & Management

1.1 Project Proposal

Project Title

Manufacturing Line Productivity and Downtime Analysis

Project Overview

This project analyzes the supplied manufacturing records to understand production-batch elapsed durations and recorded downtime patterns. The workbook contains production batch information, product reference data, downtime-factor descriptions, and downtime durations associated with batches. The team will prepare the data, build a suitable analytical model, define supported KPIs, and develop interactive Power BI dashboards.

Problem Statement

Production and downtime information can be difficult to analyze consistently when it is spread across multiple worksheets and downtime is stored in a matrix-like layout. The project will organize these records and provide a clearer view of recorded downtime by factor, product, batch, date, and operator where the data supports those comparisons. The available workbook alone does not establish actual production output, defect levels, financial losses, or the full impact of downtime.

Objectives

- Inspect, clean, and standardize the four source worksheets.
- Transform downtime data into an analysis-friendly structure and connect it to production batches and downtime-factor descriptions.
- Calculate valid batch elapsed durations and recorded-downtime indicators using the source units and documented assumptions.
- Compare recorded downtime across factors and explore differences by product, date, batch, and operator where meaningful.
- Develop six coordinated analytical dashboard pages, one per team member, integrated into a single Power BI report.
- Present evidence-based findings and practical recommendations without making claims unsupported by the data.

Scope

In scope	Out of scope unless additional data is provided
Data profiling, cleaning, transformation, and modeling of the supplied workbook	Calculating actual production volume or throughput
Batch elapsed-duration analysis and recorded downtime analysis	Defect rate, scrap rate, or quality yield
Downtime-factor comparisons, including the source Operator Error classification	Claiming that a named operator caused an incident
Product, date, batch, and operator comparisons supported by available fields	Financial cost of downtime or lost revenue
Power BI dashboards and documentation	Full OEE calculation without required planned-time, ideal-cycle, total-count, and good-count data

Expected Deliverables

- Documented and cleaned data tables.
- A documented data model and KPI definitions.
- Six Power BI dashboard pages developed and integrated by the team.
- A final set of data-supported insights and recommendations.
- Project documentation and final presentation.

1.2 Project Plan and Timeline

The plan runs over eight weeks, counted from the project kickoff date. Week numbers are relative so the plan stays valid when mapped onto the official DEPI calendar.

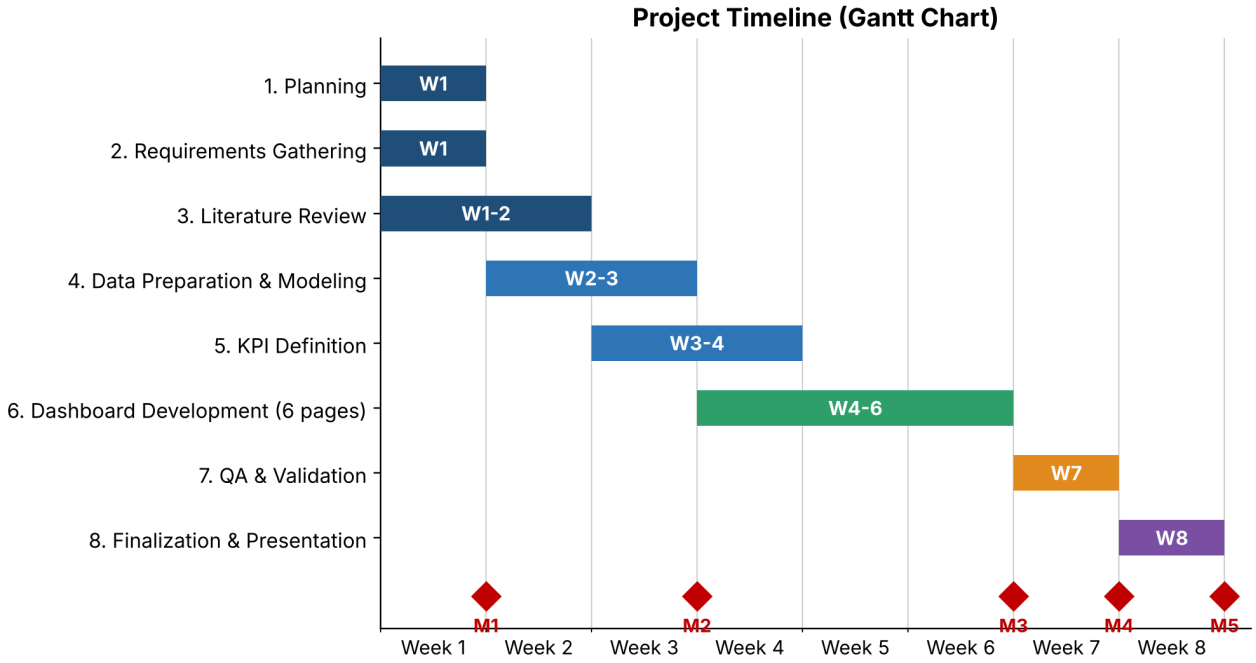


Figure 1. Project timeline (Gantt chart). Red diamonds mark milestones M1-M5, each reached at the end of the week shown.

Phase	Main tasks	Output	Timing
1. Planning	Confirm scope, objectives, roles, risks, and communication approach	Project plan	Week 1
2. Requirements Gathering	Define stakeholders, user stories, use cases, functional and non-functional requirements	Requirements specification	Week 1
3. Literature Review	Review credible sources on manufacturing productivity and downtime analysis	Literature review	Weeks 1-2
4. Data Preparation & Modeling	Profile data, clean fields, reshape downtime data, validate keys and time calculations	Prepared model and data dictionary	Weeks 2-3
5. KPI Definition	Agree KPI formulas and test them against source records	KPI catalogue	Weeks 3-4
6. Dashboard Development	Build the six individual pages and integrate them into one report	Six dashboard pages	Weeks 4-6
7. QA & Validation	Check calculations, filters, relationships, and edge cases	Testing log and corrected report	Week 7
8. Finalization	Complete documentation, recommendations, and presentation	Final submission package	Week 8

Milestones and Deliverables

Milestone	Reached at	Deliverable	Owner
M1: Scope and requirements baseline	End of Week 1	Project plan, requirements specification	Ziad (team leader), all members
M2: Data model and KPI catalogue	End of Week 3	Cleaned tables, data model, data dictionary, KPI definitions	Moddather (model); all members (cleaning and validation)
M3: Dashboards integrated	End of Week 6	Six dashboard pages merged into one Power BI report	Each member for own page; Ziad for integration
M4: QA complete	End of Week 7	Testing log, corrected report	All members; Ziad coordinates
M5: Final submission	End of Week 8	Documentation, insights, recommendations, presentation	All members

Resource Allocation

Resource	Use in the project	Allocation
Team: 6 members	Data cleaning and validation (shared); one dashboard each; documentation and presentation	About one dashboard owner per page, with all members shared on data quality
Microsoft Excel	Inspect and profile the four source worksheets	Weeks 1-3, retained as unchanged source
Power BI Desktop	Data model, DAX measures, six report pages	Weeks 2-7
Version control / shared drive	Share the model and report versions; avoid integration conflicts	Whole project
Instructor	Scope guidance and final review of the documentation and dashboards	Start and end of the project

1.3 Team Task Assignment and Responsibilities

The project work was divided among six team members. All team members collaborated on data cleaning and validation to ensure data quality and consistency. Moddather was responsible for developing the data model, and each member created the dashboard assigned to them. Ziad also served as team leader, coordinating the team's progress and ensuring consistency across the project.

Team member	Primary responsibility	Specific tasks / deliverables
Member 1: Moddather	Manufacturing Production Overview & Data Modeling	Developed the data model and relationships between tables, and created Dashboard 1 to analyze overall manufacturing production.
Member 2: Fathy	Product Performance Analysis	Created Dashboard 2 to analyze production volumes, batch processing times, product characteristics, and production efficiency.
Member 3: Alaa	Operator Performance Analysis	Created Dashboard 3 to analyze operator-related production metrics, batch processing times, and performance across products.
Member 4: Ziad (Team Leader)	Downtime Analysis	Created Dashboard 4 to analyze total downtime, downtime factors, frequency of downtime events, downtime trends, and batches with the highest downtime. Coordinated team activities and monitored consistency across the project.
Member 5: Mohamed Gamal	Production Efficiency & Optimization	Created Dashboard 5 to evaluate production efficiency, compare actual and minimum batch processing times, and identify opportunities for productivity improvement.

Team member	Primary responsibility	Specific tasks / deliverables
Member 6: Nada	Shift & Changeover Analysis	Created Dashboard 6 to analyze downtime by shift, production day, product changeovers, operator handovers, batch duration, and product size, while monitoring daily operating and idle hours.

Shared responsibilities: All team members participated in data cleaning and validation, collaborated to resolve data quality issues, and ensured that the analyses were based on consistent and reliable data.

1.4 Risk Assessment and Mitigation

Risk	Potential impact	Mitigation
Downtime units or recording rules are unclear	Incorrect totals and comparisons	Apply the source unit consistently and document whether events overlap before final KPI calculations.
An end time is earlier than its start time because a batch crosses midnight	Incorrect or negative elapsed duration	Build full datetimes and handle overnight batches explicitly; validate the affected batch manually.
Downtime data remains in a wide matrix layout	Difficult filtering and aggregation	Unpivot factor columns into rows and preserve batch/factor identifiers.
Missing output and quality fields	Some common manufacturing KPIs cannot be calculated	State limitations and request extra data if those KPIs are required.
Inconsistent definitions across the six dashboard owners	Conflicting values across pages	Use one shared model and KPI catalogue; review measures centrally.
Operator Error category is misinterpreted	Unsupported conclusions about individuals	Treat it as the source's classification of a downtime factor, not proof of individual responsibility.
Late work or integration conflicts	Delayed submission	Assign clear owners, use version control, set check-in milestones, and integrate pages early.

1.5 KPIs and Success Criteria

Data KPIs (reported in the dashboards)

KPI	Calculation concept	Caution / validation needed
Recorded Downtime	Sum of validated downtime durations	Use the source unit; summed as recorded (see section 3.8).
Downtime by Factor	Sum downtime grouped by factor description	Requires normalized downtime records and valid factor mapping.
Downtime Share (%)	Factor downtime divided by total downtime	Define behavior when the denominator is zero and clarify filter context.
Batch Elapsed Duration	End datetime minus start datetime	Handle midnight crossing; elapsed time is not necessarily machine runtime.
Batch Count	Count of distinct valid batch IDs	Check missing and duplicate IDs.
Elapsed Duration vs. Minimum Batch Time	Compare elapsed duration with product reference value	Compare only where units are compatible; the reference is the source's minimum batch time.

Project success KPIs (measure the project itself)

KPI	How it is measured	Target
Dashboard delivery	Dashboard pages completed and integrated into the single report	6 of 6 pages integrated by end of Week 6
Milestone punctuality	Milestones M1-M5 reached by their planned week	5 of 5 on time
Dashboard QA pass rate	Passed agreed test cases divided by all test cases	All critical test cases passed; no open critical defects at submission
Dashboard response time	Time for common filter and slicer interactions on a representative sample	Common filter interactions complete without noticeable delay in testing
Calculation accuracy	Sampled batches and totals reconciled manually to source data	All sampled checks match source records
User adoption / acceptance	Review at final presentation: can users answer their own questions from the report?	Users can answer their questions using the report and its filters
Documentation completeness	All DEPI sections complete, with assumptions and limitations stated	100% of required sections

Project success is judged by traceable source data, correct and documented calculations, consistent results across all six pages, usable filters, clear definitions, and recommendations supported by the analysis.

2. Literature Review

This section summarizes established concepts relevant to the project. These are general background concepts; they are applied to the team's workbook only in the sections that follow.

2.1 Manufacturing Productivity and Performance Measurement

Manufacturing performance is evaluated through indicators selected to match the production system and the question being investigated. No single metric describes every aspect of performance. For this project, the available records support analysis of batch elapsed duration and recorded downtime. They do not include production counts or ideal cycle output, so actual throughput cannot be established from this workbook alone.

2.2 Downtime Categorization and Measurement

Downtime analysis is more useful when interruptions are recorded consistently and assigned to understandable categories. A structured list of downtime factors can help users compare the duration associated with each recorded category. This supports the planned transformation of the Line downtime worksheet from a matrix-like layout into a normalized table with one row per batch and downtime factor. Duration units and event overlap are therefore documented as assumptions (section 3.8) and applied consistently when summing values.

2.3 Root-Cause Analysis and Continuous Improvement

Ranking downtime categories can help prioritize which processes or events deserve further investigation. However, a dashboard that shows an association or a high recorded duration does not, by itself, prove the root cause of a problem. The project presents descriptive findings from the supplied data and frames recommendations as areas for investigation unless additional evidence confirms a cause.

2.4 Manufacturing Monitoring and Data-Driven Decisions

Manufacturing monitoring and diagnostics use operational data to make performance patterns visible and support decisions. An interactive Power BI report can serve as a descriptive monitoring tool by allowing users to explore downtime by factor and compare available batch, product, date, and operator fields. Predictive maintenance is outside the current scope because the workbook does not provide sensor streams, maintenance histories, or an appropriate time series for reliable predictive modeling.

2.5 Relevance to This Project

Literature theme	How it informs this project	Limitations in the supplied data
Select performance measures that fit the question	Prioritize batch duration and recorded downtime measures that can be supported by source fields.	No production quantity, ideal cycle time, or good-unit count.
Use consistent downtime categories	Map factor IDs to their descriptions and normalize the downtime data.	Unit and event-overlap rules are handled as documented assumptions.
Prioritize investigation using measured losses	Rank downtime factors and compare patterns across relevant dimensions.	Duration alone does not prove root cause or lost output.
Use monitoring to support decisions	Provide interactive views with clear labels, filters, and definitions.	No sensor data or maintenance records for predictive analysis.

2.6 Literature Review Conclusion

The literature supports a disciplined approach: define indicators carefully, structure downtime records consistently, prioritize analysis based on measured patterns, and avoid claiming causation without supporting evidence. The current workbook is appropriate for an initial descriptive analysis of recorded downtime and batch elapsed duration. Additional production, quality, planned-time, ideal-cycle, maintenance, and cost data would be required to expand the analysis to throughput, yield, full OEE, predictive maintenance, or financial impact.

2.7 Suggested Improvements

These improvements would extend the project beyond the limits of the current workbook. Each one names the data or step that would be needed.

Area	Improvement	What is needed
Data definitions	Formally document the unit of downtime and minimum batch time, and whether downtime events can overlap within a batch.	A data dictionary from the source system or dataset owner.
Throughput and yield	Add output, good-unit, and defect-count measures so productivity can be measured, not only elapsed time.	Production quantity, good-unit and defect counts per batch.
Full OEE	Calculate availability, performance, and quality, and track OEE over time.	Planned production time, ideal cycle time, total and good counts.
Financial impact	Estimate the cost of recorded downtime to help prioritize action.	Cost per unit of downtime or margin per product.
Root-cause analysis	Follow up the highest-ranked downtime factors with root-cause methods such as 5 Whys or Pareto review.	Maintenance and incident records; input from line staff.
Predictive maintenance	Consider predictive models once suitable data exists.	Sensor streams, maintenance history, and a reliable time series.
Report operation	Schedule automatic data refresh and add row-level security.	Approved sharing method and a Power BI service workspace.
Usability	Test the report with real users and refine filters, labels, and layout.	A short usability session with intended users.

2.8 Grading Areas and Project Evidence

The table maps each grading area named in the DEPI guideline (documentation, implementation, testing, and presentation) to the evidence this project provides.

Grading area	Evidence in this project
Documentation	This document: planning, literature review, requirements, KPI catalogue, data dictionary, assumptions and limitations
Implementation	Cleaned data, data model, DAX measures, and the six integrated Power BI dashboard pages
Testing	QA & validation log: reconciliation to source data, filter and relationship checks, edge cases
Presentation	Final presentation of findings and recommendations

3. Requirements Gathering

3.1 Purpose

Requirements gathering defines the information needs of intended users and translates them into testable requirements for the data model and dashboard. The stakeholders below are derived from the fields and structure of the dataset.

3.2 Stakeholder Analysis

Stakeholder	Likely need	Potential dashboard support
Production Manager	Understand which recorded downtime categories account for the most time and compare production-time patterns.	Overview KPIs, downtime ranking, product/date filters.
Line Supervisor	Investigate a specific batch and inspect its recorded downtime details.	Batch-level details and downtime breakdown.
Process / Industrial Engineer	Identify patterns that may warrant root-cause investigation.	Factor comparisons and product/batch breakdowns.
Dashboard Maintainer / Data Analyst	Refresh, validate, and maintain the model and measures.	Documented transformations, data dictionary, KPI catalogue.
DEPI Instructor / Evaluator	Check that the project meets the agreed scope and documentation requirements.	Traceable requirements, test cases, assumptions, and limitations.

3.3 User Stories

ID	User story	Priority
US-01	As a production manager, I want to view recorded downtime by factor so that I can identify categories with the largest recorded durations.	High
US-02	As a line supervisor, I want to filter by date, product, batch, and operator so that I can investigate selected records.	High
US-03	As a process engineer, I want to compare batch elapsed duration with the product's minimum-batch-time reference so that I can identify records for further review.	High
US-04	As a dashboard user, I want KPI definitions and units to be visible so that I can interpret the results correctly.	High
US-05	As a dashboard maintainer, I want downtime factor IDs mapped to readable descriptions so that the report is understandable.	High
US-06	As a project evaluator, I want assumptions and limitations documented so that conclusions are not overstated.	Medium

3.4 Use Cases

UC-01: Explore Downtime by Factor

Field	Description
Primary actor	Production Manager / Process Engineer
Preconditions	Downtime records are normalized, factor descriptions are mapped, and the duration unit is confirmed.
Main flow	Open the dashboard; review total recorded downtime; select a factor or apply a date/product filter; inspect the resulting breakdown.
Alternative flow	If no records match the filters, show a clear no-data state rather than misleading results.
Postcondition	The user can inspect the selected factor's recorded duration and share of the filtered total.

UC-02: Investigate a Production Batch

Field	Description
Primary actor	Line Supervisor
Preconditions	Batch details, product mapping, timestamps, and associated downtime records have been checked.
Main flow	Select a batch and review its product, operator, date, elapsed duration, and recorded downtime factors.
Alternative flow	If no downtime is recorded for a batch, indicate 'no downtime recorded'; do not assume this proves no interruption occurred.
Postcondition	The user can review the available information for the selected batch.

3.5 Functional Requirements

ID	Requirement	Acceptance check
FR-01	The solution shall connect production batches to product reference data and downtime factor descriptions using validated keys.	Relationships are checked and unmatched keys are reported.
FR-02	The solution shall display the number of recorded batches.	The count reconciles to distinct valid batch IDs after documented checks.
FR-03	The solution shall calculate batch elapsed duration from start and end datetime values.	Overnight batches are handled correctly and checked against source records.
FR-04	The solution shall display total recorded downtime and downtime grouped by factor.	Values reconcile to the normalized downtime event table.
FR-05	The solution shall show readable downtime-factor descriptions instead of relying only on numeric IDs.	All factor IDs map to descriptions or are flagged as unmatched.
FR-06	The solution shall provide relevant filters for available date, product, batch, operator, and downtime factor.	Filters update applicable visuals consistently.
FR-07	The solution shall compare elapsed duration with the minimum batch-time reference only where the units are compatible.	The comparison is documented and uses compatible units.
FR-08	The solution shall provide definitions and units for the displayed KPIs.	Definitions are available in the report or its accompanying documentation.
FR-09	The solution shall clearly disclose limitations and avoid unsupported throughput, yield, financial-loss, or full-OEE calculations.	The report contains no unsupported KPI claims.
FR-10	The report shall contain six dashboard pages that share one data model and one KPI catalogue.	The same KPI returns the same value on every page under the same filters.

3.6 Non-functional Requirements

ID	Requirement	Verification
NFR-01 Usability	The report shall use clear page titles, consistent labels, readable units, and a consistent layout.	Review the report against an agreed usability checklist.
NFR-02 Performance	Common dashboard interactions shall complete without noticeable delay.	Test representative filter and slicer interactions.
NFR-03 Accuracy	Displayed KPIs shall match documented calculations and validated source records.	Manually reconcile sample batches and totals.
NFR-04 Reliability	Supported filters and refreshes shall not cause broken visuals or invalid calculations.	Test normal, blank, and edge-case records.
NFR-05 Maintainability	Transformations, relationships, measures, and assumptions shall be documented.	Have another team member follow the documentation.
NFR-06 Data integrity	Cleaning decisions shall be traceable and the original source data shall remain unchanged.	Retain source files and document transformations.
NFR-07 Responsible sharing	If the report is shared outside the team, access shall follow the approved sharing method.	Check access permissions and avoid unnecessary public exposure of operator names.
NFR-08 Security and confidentiality	The workbook, data model, and report shall be accessible only to the team, the instructor, and approved stakeholders. The report shall not be published to the web publicly. Where operator-level detail is not needed, operator names shall be replaced by IDs or aliases before sharing.	Review sharing settings and file permissions before submission; confirm that shared copies contain no unnecessary personal identifiers.

3.7 Data Requirements and Constraints

- Use the batch identifier to connect production records to downtime records after validating uniqueness and completeness.
- Use the product code to connect production records to product reference data.
- Use downtime-factor ID to connect downtime event rows to factor descriptions and the source Operator Error flag.
- Reshape the downtime worksheet from a wide matrix into a long table with fields such as Batch ID, Downtime Factor ID, and Downtime Duration.
- Construct complete start and end datetimes and handle overnight batches so that elapsed duration is never negative.
- Use the units recorded in the source for downtime and minimum batch time, and compare or form ratios only where units are compatible.
- The workbook does not provide production quantity, good-unit count, defect count, planned production time, ideal cycle time, or downtime cost. Therefore, throughput, yield, full OEE, and financial loss are not supported by the current data alone.
- Treat Operator Error as a source classification attached to a downtime factor, not proof that a particular named operator caused an event.

3.8 Assumptions

Assumption	How it is handled
Downtime duration and minimum batch time are expressed in the units recorded in the source workbook.	Values are reported in the source units; comparisons are made only where the units are compatible.
Downtime records are summed as recorded.	Totals reconcile to the normalized downtime event table; overlap is not adjusted.
Batches that cross midnight are valid records.	Full datetimes are built so that elapsed duration is never negative.
Operator Error is a source classification of a downtime factor.	It is reported as a classification and is not attributed to named individuals.
The stakeholders in section 3.2 represent the likely users of the report.	Dashboard content and filters are designed around their needs.
The workbook is descriptive data only.	No throughput, yield, financial-loss, full-OEE, or predictive claims are made.

3.9 Requirements Baseline

These requirements are the baseline that all six team members work from. They cover the scope, KPI definitions, data assumptions, dashboard-page allocation, and acceptance tests, and any later change is applied consistently across the six dashboard pages.